



Parameter estimation and reliable fault detection of electric motors

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Abstract:

Accurate model identification and fault detection are necessary for reliable motor control. Motor-characterizing parameters experience substantial changes due to aging, motor operating conditions, and faults. Consequently, motor parameters must be estimated accurately and reliably during operation. Based on enhanced model structures of electric motors that accommodate both normal and faulty modes, this paper introduces bias-corrected least-squares (LS) estimation algorithms that incorporate functions for correcting estimation bias, forgetting factors for capturing sudden faults, and recursive structures for efficient real-time implementation. Permanent magnet motors are used as a benchmark type for concrete algorithm development and evaluation. Algorithms are presented, their properties are established, and their accuracy and robustness are evaluated by simulation case studies under both normal operations and inter-turn winding faults. Implementation issues from different motor control schemes are also discussed.

Keywords: Electric machine; Parameter estimation; Fault detection; Brushless direct current (BLDC) motor; Bias correction; Forgetting factor

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1 Introduction

This paper introduces new methods for accurate parameter estimation and reliable fault detection of inverter powered electric motors. Brushless direct current (BLDC) and permanent magnet alternate current (PMAC) motors are used as a benchmark platform to develop our methods. Electric motors are essential parts

of electric and hybrid vehicle powertrains [1–3] and other diversified industrial applications [4]. Accurate model identification and fault detection are necessary for reliable motor control [5]. Motor-characterizing parameters experience substantial changes or sudden jumps due to aging, motor operating conditions, or faults [6, 7]. Consequently, motor parameters must be estimated ac-

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curately during operation, leading to a system identification problem [8, 9].

This paper employs 3-phase motors as a platform to develop algorithms for identifying motor parameters during normal operations and detecting stator winding faults. To facilitate this study, an enhanced model of 3-phase permanent magnet (PM) motors is developed that accommodates both normal and faulty operating conditions. Due to high measurement noise, motor parameter estimation is a challenging problem. Both motor inputs and outputs are corrupted by noise, leading to an errors-in-variables identification (EIV) problem [10]. An EIV structure is known to introduce identification bias [11]. Motor faults entail sudden jumps in motor dynamics. To diagnose the faults promptly, identification algorithms must achieve a good balance between fast fault detection (which prefers a short data window), and noise attenuation (which is achieved by averaging, preferably over a large data window). Also, motor controller frameworks are pre-designed and must be accommodated in system identification.

This paper introduces an enhanced least-squares (LS) estimation algorithm that incorporates a function for correcting estimation bias, a forgetting factor for capturing sudden faults, and a recursive structure for efficient real-time implementation. Algorithms are presented, their properties are established, and their accuracy and robustness are evaluated by simulation case studies under both normal operations and inter-turn winding faults. One contribution of this paper is the development of new bias correction algorithms with forgetting factors in a recursive structure. Traditionally, bias correction in an EIV problem was treated by modified correction terms, instrumental methods, or prediction error methods [10–12]. The issues of estimation bias and its correction for battery model identification and state-of-charge estimation were discussed in our earlier papers [13, 14], without consideration of forgetting factors and the corresponding recursive algorithms.

The rest of the paper is organized into the following sections. Section 2 establishes enhanced model structures for three-phase balanced PM motors in normal and faulty conditions. Identification algorithms are introduced in Section 3. In Section 4, bias-corrected LS algorithms are presented and their bias correction capabilities are established. Section 5 discusses practical aspects of motor estimation, involving different motor

control schemes. Reliability of parameter estimation under these schemes is studied. For fast diagnosis of faults, system identification must balance speed and accuracy. Section 6 introduces forgetting factors into our bias correction algorithms. Recursive algorithms are derived. Section 7 concentrates on inter-turn fault diagnosis. Basic algorithms are introduced and evaluated by case studies. Section 8 highlights the main findings of this paper and points out some worthy open problems. Some preliminary ideas of this paper were reported in [15].

2 Enhanced PM motor models under normal operation and faulty conditions

This section describes models of surface mounted PM motors. For working principles, types, mechanisms, and control systems of PM motors, we refer the reader to [4, 6] for details. Exploration on modeling and diagnosis of surface mounted PM machines can be found in [5, 7, 16–18]. In this paper, we introduce an enhanced model for PM motors in normal and faulty conditions. The three-phase balanced stator windings under normal operating conditions are illustrated in Fig. 1. Under a balanced construction, all phases have the same parameters and are symmetric.

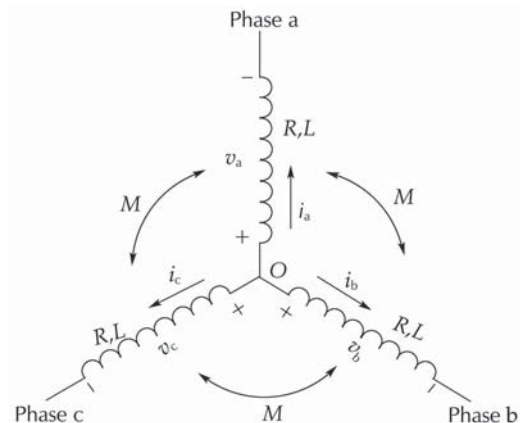


Fig. 1 Three phase stator winding.

We start with models of healthy stator windings; see Fig. 1, in which the windings are assumed to be sinusoidally distributed.¹ Since the stator windings are balanced, without loss of generality, we use phase a as a generic phase. The state equation for healthy stator windings is

$$v_a = Ri_a + \frac{d\lambda_a}{dt}, \quad (1)$$

¹ It should be emphasized that the model structures are also valid under other types of flux linkages and back electromotive force (EMF), such as trapezoidal types.

where v_a is the phase-a winding terminal voltage (V), i_a is the phase-a current (A), R is the phase-a resistance (Ω), and λ_a is the total phase-a flux linkage (Wb). Under the assumption of magnetic linearity and infinite permeability of iron, the flux linkage is related to the phase current and magnetic coupling by $\lambda_a = Li_a + Mi_b + Mi_c + \lambda_M \cos(2\pi ft + \delta_a)$. Here, L is the phase-a inductance (H), M is the stator phase cross-inductance (H), λ_M is the stator/rotor magnetic coupling flux linkage (Wb), f is the electric angular speed of the rotor (Hz), and typically $\delta_a = 0$ (rad) ($\delta_b = -2\pi/3$ and $\delta_c = 2\pi/3$).

Assume that there is no saliency, i.e., the air gap between the rotor and the stator is constant. Then, the stator inductance is constant and does not depend on the relative rotor position. It follows from (1) that

$$v_a = Ri_a + L \frac{di_a}{dt} + M \frac{di_b}{dt} + M \frac{di_c}{dt} - \lambda_M 2\pi f \sin(2\pi ft + \delta_a). \quad (2)$$

This can be written compactly as

$$v(t) = RLi(t) + H \frac{di(t)}{dt} + \lambda_M g(t), \quad (3)$$

where

$$v(t) = [v_a(t) \ v_b(t) \ v_c(t)]', \quad i(t) = [i_a(t) \ i_b(t) \ i_c(t)]',$$

$$H = \begin{bmatrix} L & M & M \\ M & L & M \\ M & M & L \end{bmatrix}, \quad g(t) = [g_a(t) \ g_b(t) \ g_c(t)]'$$

with

$$g_a(t) = -2\pi f \sin(2\pi ft + \delta_a),$$

$$g_b(t) = -2\pi f \sin(2\pi ft + \delta_b),$$

$$g_c(t) = -2\pi f \sin(2\pi ft + \delta_c).$$

Within this frame, when the stator is subject to a winding fault, model (3) is perturbed. We will use the phase-a fault as a benchmark case in our derivations, see Fig. 2. Detection algorithms for phase-b and phase-c faults are similar.

Suppose that the original number of turns of phase a is N_a for which N_{as} turns are shorted. Denote $\mu = N_{as}/N_a$, the ratio of faulty turns. It is noted that the fault introduces a fault current i_f through the bypass branch of resistance R_f in Fig. 2. Fault diagnosis is built on the following enhanced model which captures inter-turn faults with bypass resistance. Here, we assume that the healthy motor model has been identified with model pa-

rameters R, L, M, λ_M estimated. Fault detection aims to identify additional parameters that represent inter-turn faults. From Fig. 2, such parameters include μ and R_f .

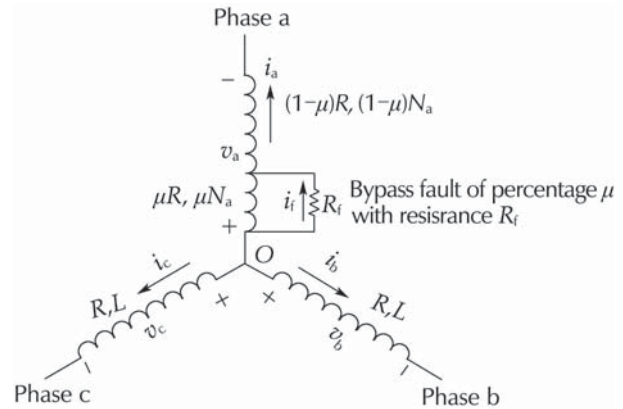


Fig. 2 Three phase stator windings with a bypass fault in phase a.

Following the same principles as before, under a fault of $0 < \mu < 1$ in phase a with resistance R_f , the model (3) is perturbed to

$$v_a = v_f + (1 - \mu)Ri_a + \frac{d}{dt}((1 - \mu)^2 Li_a + (1 - \mu)Mi_b + (1 - \mu)\mu L(i_a - i_f) + (1 - \mu)Mi_c) + (1 - \mu)e_a,$$

$$v_f = \mu R(i_a - i_f) + \frac{d}{dt}(\mu(1 - \mu)Li_a + \mu^2 L(i_a - i_f) + \mu Mi_b + \mu Mi_c) + \mu e_a,$$

$$v_b = Ri_b + \frac{d}{dt}((1 - \mu)Mi_a + \mu M(i_a - i_f) + Li_b + Mi_c) + e_b,$$

$$v_c = Ri_c + \frac{d}{dt}((1 - \mu)Mi_a + \mu M(i_a - i_f) + Mi_b + Li_c) + e_c,$$

$$v_f = R_f i_f,$$

where v_f is the voltage cross the faulty turns. By eliminating v_f , we obtain

$$\begin{cases} v_a = Ri_a - \mu Ri_f + \frac{d}{dt}(Li_a + Mi_b + Mi_c - \mu Li_f) + e_a, \\ v_b = Ri_b + \frac{d}{dt}(Mi_a + Li_b + Mi_c - \mu Mi_f) + e_b, \\ v_c = Ri_c + \frac{d}{dt}(Mi_a + Mi_b + Li_c - \mu Mi_f) + e_c, \\ R_f i_f = \mu R(i_a - i_f) + \frac{d}{dt}(\mu Li_a + \mu Mi_b + \mu Mi_c - \mu^2 Li_f) + \mu e_a. \end{cases} \quad (4)$$

The first equation implies

$$\mu v_a = \mu Ri_a - \mu^2 Ri_f + \frac{d}{dt}(\mu Li_a + \mu Mi_b + \mu Mi_c - \mu^2 Li_f) + \mu e_a,$$

which, after substituting into the fourth equation, leads to

$$i_f = \frac{\mu}{R_f + \mu R - \mu^2 R} v_a, \quad v_f = \frac{\mu R_f}{R_f + \mu R - \mu^2 R} v_a. \quad (5)$$

It is interesting to note that this relationship between v_f and v_a is independent of the value L .

It follows that $\mu i_f = \frac{\mu^2}{R_f + \mu R - \mu^2 R} v_a = \kappa v_a$ where

$$\kappa = \frac{\mu^2}{R_f + \mu R - \mu^2 R}. \quad (6)$$

Now, using (6) to eliminate i_f in the first three equations in (4) results in

$$\begin{aligned} v_a &= Ri_a - \kappa R v_a + \frac{d}{dt}(Li_a + Mi_b + Mi_c - \kappa L v_a) + e_a, \\ v_b &= Ri_b + \frac{d}{dt}(Mi_a + Li_b + Mi_c - \kappa M v_a) + e_b, \\ v_c &= Ri_c + \frac{d}{dt}(Mi_a + Mi_b + Li_c - \kappa M v_a) + e_c. \end{aligned}$$

These can be compactly expressed as

$$v = RLi - \kappa G_1 v_a + H \frac{di}{dt} - \kappa G_2 \frac{dv_a}{dt} + g, \quad (7)$$

where H and g are defined before, and $G_1 = (R, 0, 0)'$, $G_2 = (L, M, M)'$.

Next, we discretize (7) for implementation of algorithms on a computer. Suppose that the sampling interval is τ . Let $v_k = v(k\tau)$, $i_k = i(k\tau)$, $g_k = g(k\tau)$. Then, (7) is discretized to

$$\begin{aligned} i_{k+1} &= (I - \tau RH^{-1})i_k + H^{-1} \\ &\times \begin{bmatrix} \tau(v_a(k) - e_a(k)) + \kappa\mu(\tau R v_a(k) + L(v_a(k) - v_a(k-1))) \\ \tau(v_b(k) - e_b(k)) + \kappa\mu M(v_a(k) - v_a(k-1)) \\ \tau(v_c(k) - e_c(k)) + \kappa\mu M(v_a(k) - v_a(k-1)) \end{bmatrix}, \end{aligned} \quad (8)$$

where we have

$$\begin{bmatrix} e_a(k) \\ e_b(k) \\ e_c(k) \end{bmatrix} = \lambda_M \begin{bmatrix} g_a(k) \\ g_b(k) \\ g_c(k) \end{bmatrix} \equiv \lambda_M g(t).$$

This paper investigates motor parameter estimation under normal operating conditions and fault detection. Simulation models will be used to schedule a fault appearance. Starting with a normal operation, a fault is

then simulated in phase a at a certain time. Our enhanced model is then used to represent the voltage-current profiles after the fault. These will be covered in the subsequent sections.

3 Identification algorithms

3.1 Regression models for system identification

The healthy motor model (3) contains four parameters R , L , M and λ_M . For system identification, we rewrite (3) in the form of

$$v(t) = Ri(t) + (LI + MH_M) \frac{di(t)}{dt} + \lambda_M g(t), \quad (9)$$

where I is the identity matrix and $H_M = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$. Under

the sampling interval τ , (9) is discretized to

$$v_k \tau = Ri_k \tau + L(i_k - i_{k-1}) + MH_M(i_k - i_{k-1}) + \lambda_M g_k \tau. \quad (10)$$

Denote $\theta = [R \ L \ M \ \lambda_M]'$, $\phi'_k = [i_k \tau \ i_k - i_{k-1} \ H_M(i_k - i_{k-1}) \ g_k \tau]'$; $y_k = v_k \tau$. It follows that (10) can be written in a regression form

$$y_k = \phi'_k \theta. \quad (11)$$

It is noted that the dimensions are $y_k \in \mathbb{R}^3$, $\phi'_k \in \mathbb{R}^{3 \times 4}$, $\theta \in \mathbb{R}^4$. Also, although physically it is more convenient to view the phase voltages as the input and the currents as the output for the motor models, for system identification we follow (11) to view v_k as the output and i_k as the input. As a result, in the subsequent discussions, output noises will refer to voltage measurement noises and input noises will be current measurement noises.

3.2 Algorithms

Due to measurement errors and disturbances, observations are corrupted by noises $\tilde{y}_k = y_k + e_k$; $\tilde{i}_k = i_k + \varepsilon_k$. Current measurement noises introduce a perturbation on the regressor $\tilde{\phi}'_k = \phi'_k + \delta_k$. Consequently, the regression relationship that utilizes measured values is $\tilde{y}_k = (\tilde{\phi}'_k - \delta_k)\theta + e_k$. Here, e_k is due to noises on the voltage and δ_k is induced from the current measurement noises.

Assumption 1 The joint vector sequence $\{[\varepsilon_k, e_k]\}$ is stationary and strongly ergodic (in the sense of convergence with probability one (w.p.1)) such that

$E[\varepsilon_k, e_k]' = 0$, $E(\|[\varepsilon_k, e_k]\|^2) < \infty$, and that both $\{[\varepsilon_k, e_k]\}$ and $\{[\varepsilon_k, e_k]'[\varepsilon_k, e_k]\}$ are ergodic. That is, $\frac{1}{N} \sum_{k=1}^N [\varepsilon_k, e_k] \rightarrow 0$, w.p.1 as $N \rightarrow \infty$, $\frac{1}{N} \sum_{k=1}^N [\varepsilon_k, e_k][\varepsilon_k, e_k]' \rightarrow S$, where S is a nonnegative definite matrix w.p.1 as $N \rightarrow \infty$. Here, $E(\cdot)$ denotes the expectation.

Note that the noises are zero mean, but we do not need the sequences $\{\varepsilon_k\}$ and $\{e_k\}$ to be independent or uncorrelated. A sufficient condition to ensure the ergodicity in the above assumption is that the underlying sequence is a stationary φ -mixing sequence, which is a sequence whose remote past and distant future are asymptotically independent. The well-known results [19, p. 488] then yield that $[\varepsilon_k, e_k]$ and $\{[\varepsilon_k, e_k][\varepsilon_k, e_k]'\}$ are strongly ergodic.

After N observations, denote

$$\begin{aligned}\tilde{Y}_N &= [\tilde{y}_1 \ \tilde{y}_2 \ \cdots \ \tilde{y}_N]', \quad Y_N = [y_1 \ y_2 \ \cdots \ y_N]', \\ \tilde{\Phi}_N &= [\tilde{\phi}_1 \ \tilde{\phi}_2 \ \cdots \ \tilde{\phi}_N]', \quad \Phi_N = [\phi_1 \ \phi_2 \ \cdots \ \phi_N]', \\ \Delta_N &= [\delta_1 \ \delta_2 \ \cdots \ \delta_N]'. \end{aligned}$$

Then, $\tilde{Y}_N = Y_N + E_N$; $\tilde{\Phi}_N = \Phi_N + \Delta_N$. From $\tilde{y}_k = (\tilde{\phi}_k' - \delta_k)\theta + e_k$, the observation equation becomes $\tilde{Y}_N = (\tilde{\Phi}_N - \Delta_N)\theta + E_N$. When $\tilde{\Phi}_N' \tilde{\Phi}_N$ is nonsingular, w.p.1, the standard LS estimate is

$$\theta_N = (\tilde{\Phi}_N' \tilde{\Phi}_N)^{-1} \tilde{\Phi}_N' \tilde{Y}_N = \left(\frac{1}{N} \tilde{\Phi}_N' \tilde{\Phi}_N \right)^{-1} \frac{1}{N} \tilde{\Phi}_N' \tilde{Y}_N. \quad (12)$$

We illustrate our basic algorithms with the following example.

Example 2 A PM motor has the following true parameters: $R = 2.8750 \ \Omega$, $L = 0.0064 \ \text{H}$, $M = -0.0021 \ \text{H}$, $\lambda_M = 0.1750 \ \text{Wb}$. This model is simulated on a Matlab platform. The sampling frequency is 100 kHz or equivalently the sampling interval is $\tau = 0.01 \ \text{ms}$. The applied voltage profiles are balanced three-phase sinusoid waveforms of peak value 500 V and frequency 60 Hz. The simulation is run for a total 2000 sampling points. The output (voltage) is corrupted by noise, which is a Gaussian i.i.d. (independent and identically distributed) process of zero mean and standard deviation $\sigma_v = 20 \ \text{V}$. The LS algorithm (12) is applied. Fig. 3 demonstrates the parameter estimation error trajectories. The error is defined as $\|\theta_N - \theta\|$ where $\|\cdot\|$ is the Euclidean norm. In this case, estimation is quite accurate.

We demonstrate in Section 4 that if the input is also subject to measurement noise, this algorithm will introduce identification bias, namely, parameter estimates

will converge to values different from the true value.

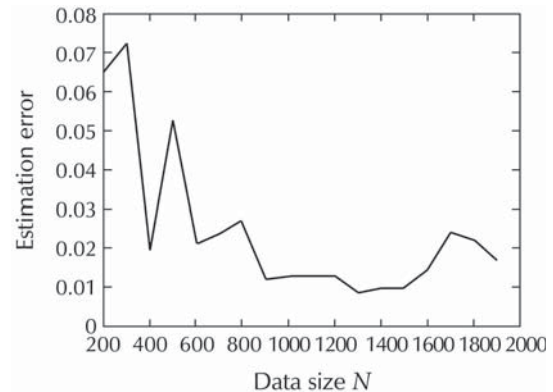


Fig. 3 Estimation error trajectories with output noise only.

4 Identification bias and correction

4.1 Errors-in-variables identification and estimation bias

Under Assumption 1, since Y_N and Φ_N are deterministic, as $N \rightarrow \infty$, with probability one, $\frac{1}{N} \Phi_N' Y_N \rightarrow A$; $\frac{1}{N} \Delta_N' Y_N \rightarrow 0$; $\frac{1}{N} \Phi_N' E_N \rightarrow 0$; $\frac{1}{N} \Delta_N' E_N \rightarrow B$; $\frac{1}{N} \Phi_N' \Phi_N \rightarrow C$, $\frac{1}{N} \Delta_N' \Delta_N \rightarrow \Sigma$; $\frac{1}{N} \Phi_N' \Delta_N \rightarrow 0$, which imply $\frac{1}{N} \tilde{\Phi}_N' \tilde{Y}_N \rightarrow A + B$ and $\frac{1}{N} \tilde{\Phi}_N' \tilde{\Phi}_N \rightarrow C + \Sigma$, for some matrices A , B , C and Σ . As a result, $\theta_N \rightarrow (C + \Sigma)^{-1}(A + B)$, w.p.1. On the other hand, the true parameter θ satisfies $\theta = C^{-1}A$. Consequently, we have the following theorem.

Theorem 3 Assume that Assumption 1 holds and that C^{-1} exists. Then, the least-squares estimate (12) is asymptotically biased in that $\lim_{N \rightarrow \infty} (\theta_N - \theta) = (C + \Sigma)^{-1}(B - \Sigma\theta)$ w.p.1.

Proof This follows from

$$\begin{aligned}\theta_N - \theta &\rightarrow (C + \Sigma)^{-1}(A + B) - C^{-1}A \\ &= (C + \Sigma)^{-1}A - C^{-1}A + (C + \Sigma)^{-1}B \\ &= -(C + \Sigma)^{-1}\Sigma C^{-1}A + (C + \Sigma)^{-1}B \\ &= (C + \Sigma)^{-1}(B - \Sigma\theta).\end{aligned}$$

This completes the proof.

Example 4 To demonstrate the impact of current measurement noises, we consider the same motor as in Example 2. Fig. 4 compares two cases: 1) Only voltage measurements have noises; 2) both voltage and current measurements are subject to noises. The same least-squares algorithm (12) is applied to both cases. In the

first case, only the output (voltage) is corrupted by noise, which is a sequence of Gaussian i.i.d. random variables with zero mean and standard deviation $\sigma_v = 20$ V. Since the voltage peak is 500 V, or equivalently root mean square (RMS) value $500/\sqrt{2} = 353.6$, this amounts to a noise-to-signal ratio of 5.66%. The top plot shows that when no input noise exists, the LS algorithm is quite effective in generating reliable parameter estimates. Then, an input noise is added to the current measurements, which is a Gaussian i.i.d. sequence with zero mean and standard deviation $\sigma_i = 5$ A. Since the current magnitudes are close to 116 A (or 82 RMS), this is about a noise-to-signal ratio of 6.1%. The bottom plot illustrates that the parameter estimation now has a bias, which is about 2.486. Since the size of the parameter vector $\|\theta\|$ is 2.8803, this bias is a relative estimation error 86.3% which is obviously unacceptable. It is noted that this is a persistent bias that does not decrease with an increase in data size. In comparison, without the input noise, the estimation error is below 0.02 or around 0.7%.

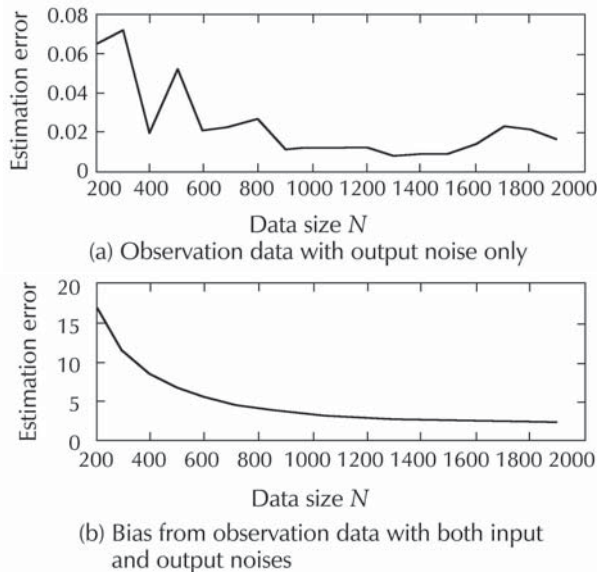


Fig. 4 Impact of input measurement noise on estimation bias.

4.2 Bias correction by modified LS algorithms

Identification bias can be corrected if Σ and B are known. Algorithm (12) is now modified to

$$\theta_N = \left(\frac{1}{N} \tilde{\Phi}'_N \tilde{\Phi}_N - \Sigma \right)^{-1} \left(\frac{1}{N} \tilde{\Phi}'_N \tilde{Y}_N - B \right). \quad (13)$$

If Σ and B are unknown, we can use statistical methods to estimate them. Then, in the bias correction algorithm (13), in place of the true Σ and B , we can use their

estimates.

Theorem 5 Under the assumptions of Theorem 3, the estimates in (13) satisfy $\theta_N \rightarrow \theta$, w.p.1 as $N \rightarrow \infty$.

Proof By the strong law of large numbers, as $N \rightarrow \infty$,

$$\theta_N \rightarrow (C + \Sigma - \Sigma)^{-1}(A + B - B) = C^{-1}A = \theta \text{ w.p.1.}$$

This completes the proof.

The modified LS algorithm (13) can be recursified for real-time computational efficiency. The following recursive algorithm was introduced in [13].

Theorem 6 [13] The estimates θ_N in (13) can be updated recursively as

$$\begin{aligned} \theta_N &= \theta_{N-1} + K_N \left(\begin{bmatrix} \tilde{y}_N \\ B \end{bmatrix} - \begin{bmatrix} \tilde{\phi}'_N \\ \Sigma \end{bmatrix} \theta_{N-1} \right), \\ K_N &= P_{N-1} [\tilde{\phi}_N - I] (I + \begin{bmatrix} \tilde{\phi}'_N \\ \Sigma \end{bmatrix} P_{N-1} [\tilde{\phi}_N - I])^{-1}, \\ P_N &= P_{N-1} - K_N \begin{bmatrix} \tilde{\phi}'_N \\ \Sigma \end{bmatrix} P_{N-1}. \end{aligned}$$

Example 7 Continuing the study from Example 4, we note that when the input noise exists and bias correction method is not applied, at the exit point ($N = 2000$) the norm of the estimation error is 2.486 (a sample result in simulation). Now we apply our bias correction algorithm, the estimation error at the exit point is reduced to 0.0061.

5 Case studies on parameter estimation and implementation considerations

Practical motors involve certain physical system structures, nonlinearities, auxiliary driving circuits, and time delays. This section includes some more realistic simulation studies that accommodate further motor details.

Two types of stator construction are common for PM machines: sinusoidal winding distribution for permanent magnet synchronous machines (PMSM) and concentrated winding for BLDC motors. In the first case, the back EMF is sinusoidal. The back EMF under concentrated winding is trapezoidal. One important difference between these two types is that synchronous machines have continuous currents through all windings (180-degree current leads). In contrast, BLDC machines will have 'square' currents with 120-degree leads. Consequently, for each winding there is a time interval when

there is no current through a particular winding.

Typical PM configurations include the six-step controlled PM motor shown in Fig. 5, the field-oriented control (FOC), and the self-controlled system. In the six-step motor, its inverter has six signal levels and requires the lowest closed-loop bandwidth. Sensor delay is a critical parameter because it results in model mismatch. The FOC motor directly controls the stator rotating magnetic field on the rotating frame to provide maximal torque generation and to ensure smoothness of rotor movements. The self-controlled operation is a simplified FOC that employs a stator-based coordinate frame. It is simple in construction, but requires high

bandwidths, generates more noise, and is less smooth in rotor movements than the other two types [18].

We now present simulation studies for stator winding parameter identification under normal operating conditions. The motor is a six-step controlled motor with sinusoid state winding. The motor true parameters are the same as in Example 2. In this case, R , L , M and λ_M are to be estimated under a closed-loop configuration. The model sampling time is 0.1 ms. A total of 10000 data points are used in this study. Due to PWM control circuits, the driving voltages' profiles are no longer sinusoid waveforms. The phase current waveforms are also quite different. These are shown in Fig. 6.

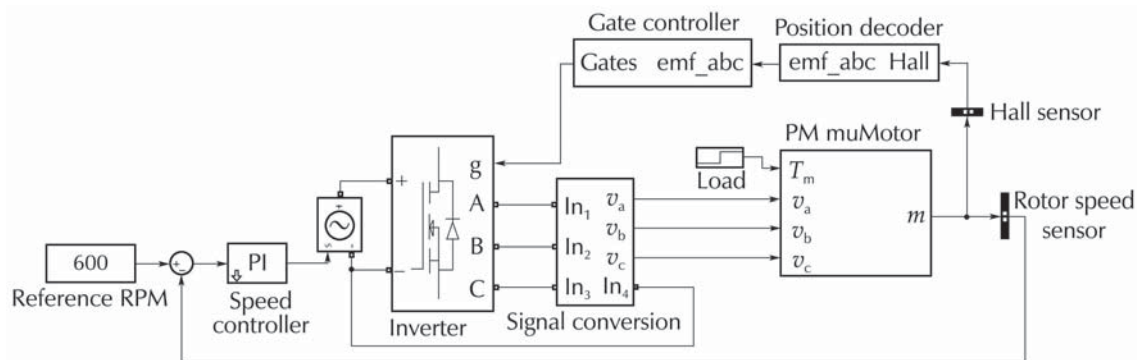


Fig. 5 Six-step controlled PM motor.

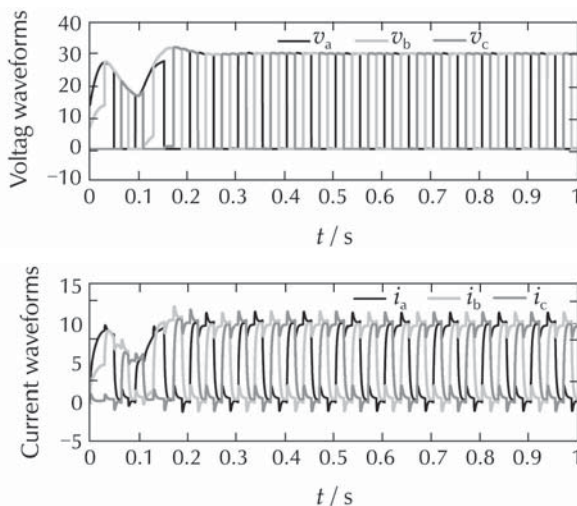
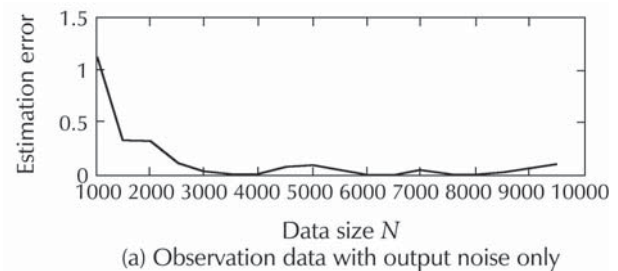


Fig. 6 Phase voltage and current profiles.

To understand further the impact of current measurement noises, we compare two cases: 1) Only voltage (output in system identification) measurements have noises; 2) both voltage and current measurements are subject to noises. The least-squares algorithm (12) is

applied. In the first case, only the output (voltage) is corrupted by noise, which is a sequence of Gaussian i.i.d. random variables with zero mean and standard deviation $\sigma_v = 50$ V. Estimates are shown in Fig. 7. The top plot shows that when no input noise exists, the LS algorithm generates highly accurate estimates. When an input noise is added to the current measurements, which is a Gaussian i.i.d. sequence with zero mean and standard deviation $\sigma_i = 10$ A, the bottom plot illustrates that the parameter estimation has a bias, which is about 2.1, or a relative estimation error 72%. This is a persistent bias that does not decrease with an increase in data size.



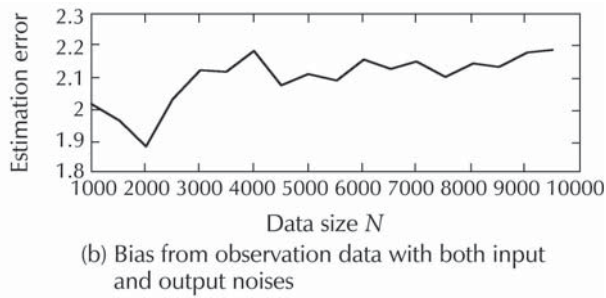


Fig. 7 Impact of input measurement noise on estimation bias.

The bias correction algorithm (13) is then applied. The estimated parameter values at the exit point are listed in Table 1. The norm of the estimation error is 0.011, or a relative error 0.3826%.

Table 1 Estimates from bias-corrected LS algorithm.

	R	L	M	λ_M
True values	2.875	0.0064	− 0.0021	0.1750
Estimates	2.8720	0.0063	− 0.0022	0.1644
Estimation errors	0.003	0.0001	− 0.0001	0.0106

6 Fast tracking and forgetting factors

Faults are sudden changes in motor model structures and parameters. When a fault occurs, the identification algorithms must detect such changes accurately and quickly. If large historical data are used in system identification, the new data that reflect the fault will have small impact on the overall parameter estimation, leading to a very slow detection process. The standard LS algorithm minimizes $\min_{\theta} \sum_{k=0}^N (y_k - \phi'_k \theta)^2$ in which all data are equally weighted.

One useful technique to overcome this drawback is to discard old data in a systematic way so that the new data will have more weight in the identification process. This can be achieved by adding a forgetting factor λ in the least squares criterion: for $0 < \lambda < 1$, we modify the optimization problem to $\min_{\theta} \sum_{k=0}^N \lambda^{N-k} (y_k - \phi'_k \theta)^2$, which exponentially weights down old data. In its matrix form, this is equivalent to

$$\min_{\theta} (Y_N - \Phi_N \theta)' W_N (Y_N - \Phi_N \theta), \quad (14)$$

where $W_N = \text{diag}\{\lambda^N, \lambda^{N-1}, \dots, \lambda, 1\}$. When $\lambda = 1$, it is reduced to the un-weighted LS algorithm. When λ is close to 0, only most recent data are used in estimating

parameters. λ is called a ‘forgetting factor’. There is a key trade-off in selecting λ . If λ is close to 1, then historical data remain heavily weighted. Consequently, fault detection will be slow. On the other hand, if λ is small, the fault detection will be faster, but noise attenuation capability will be compromised, which follows from the laws of large numbers [19].

Let $Q_N = W_N^{1/2}$. Then, (14) can be written as

$$\begin{aligned} & \min_{\theta} (Y_N - \Phi_N \theta)' W_N (Y_N - \Phi_N \theta) \\ & = \min_{\theta} (Q_N Y_N - Q_N \Phi_N \theta)' (Q_N Y_N - Q_N \Phi_N \theta), \end{aligned}$$

whose solution can be obtained by the LS result with Y_N replaced by $Q_N Y_N$ and Φ_N by $Q_N \Phi_N$, as

$$\theta_N = (\Phi_N' W_N \Phi_N)^{-1} \Phi_N' W_N Y_N. \quad (15)$$

When both input and output noises are taken into consideration, (15) becomes

$$\theta_N = (\tilde{\Phi}_N' W_N \tilde{\Phi}_N)^{-1} \tilde{\Phi}_N' W_N \tilde{Y}_N. \quad (16)$$

However, when input noises cause bias in LS estimates, (16) will be subject to bias as well. We note that $Y_N = \Phi_N \theta$, which implies $Q_N Y_N = Q_N \Phi_N \theta$. As a result, $\theta = (\Phi_N' W_N \Phi_N)^{-1} \Phi_N' W_N Y_N$. On the other hand,

$$\begin{aligned} \theta_N &= ((\Phi_N + \Delta_N)' W_N (\Phi_N + \Delta_N))^{-1} \\ &\quad \times (\Phi_N + \Delta_N)' W_N (Y_N + E_N). \end{aligned}$$

Under Assumption 1, $E(\Phi_N' W_N E_N) = 0$, $E(\Delta_N' W_N Y_N) = 0$, $E(\Delta_N' W_N \Phi_N) = 0$. Denote $E(\Delta_N' W_N \Delta_N) = \Sigma_N$, $E(\Delta_N' W_N E_N) = B_N$. Then, the modified LS estimation (for bias correction) with forgetting factor is given by

$$\theta_N = (\tilde{\Phi}_N' W_N \tilde{\Phi}_N - \Sigma_N)^{-1} (\tilde{\Phi}_N' W_N \tilde{Y}_N - B_N). \quad (17)$$

We should point out that since $0 < \lambda < 1$, the factor $1/N$ in (13) is no longer needed here. We now derive a recursive algorithm for (17). Let $\Sigma = E(\delta_N' \delta_N)$ and $B = E(\delta_N' e_N)$. By stationarity, these quantities do not depend on N .

Theorem 8 Given a forgetting factor $0 < \lambda \leq 1$, the bias-corrected LS estimate θ_N with forgetting factor λ in (17) can be updated recursively as

$$\begin{aligned} \theta_N &= \theta_{N-1} + K_N \left(\begin{bmatrix} \tilde{y}_N \\ B \end{bmatrix} - \begin{bmatrix} \tilde{\Phi}_N' \\ \Sigma \end{bmatrix} \theta_{N-1} \right), \\ K_N &= P_{N-1} [\tilde{\Phi}_N, -I] (\lambda I + \begin{bmatrix} \tilde{\Phi}_N' \\ \Sigma \end{bmatrix} P_{N-1} [\tilde{\Phi}_N, -I])^{-1}, \end{aligned}$$

$$P_N = \frac{P_{N-1}}{\lambda} - K_N \begin{bmatrix} \tilde{\phi}'_N \\ \Sigma \end{bmatrix} \frac{P_{N-1}}{\lambda}.$$

Proof From (17), $\theta_N = (\tilde{\Phi}'_N W_N \tilde{\Phi}_N - \Sigma_N)^{-1} (\tilde{\Phi}'_N W_N \tilde{Y}_N - B_N)$. On the other hand,

$$\begin{aligned} \tilde{\Phi}'_N W_N \tilde{\Phi}_N &= \lambda \tilde{\Phi}'_{N-1} W_{N-1} \tilde{\Phi}_{N-1} + \tilde{\phi}_N \tilde{\phi}'_N, \\ \Sigma_N &= \lambda \Sigma_{N-1} + \Sigma, \\ \tilde{\Phi}'_N W_N \tilde{Y}_N &= \lambda \tilde{\Phi}'_{N-1} W_{N-1} \tilde{Y}_{N-1} + \tilde{\phi}'_N \tilde{y}_N, \\ B_N &= \lambda B_{N-1} + B. \end{aligned}$$

Let $P_N = (\tilde{\Phi}'_N W_N \tilde{\Phi}_N - \Sigma_N)^{-1}$. Then,

$$\begin{aligned} P_N &= (\tilde{\Phi}'_N W_N \tilde{\Phi}_N - \Sigma_N)^{-1} \\ &= (\lambda \tilde{\Phi}'_{N-1} W_{N-1} \tilde{\Phi}_{N-1} + \tilde{\phi}_N \tilde{\phi}'_N - \lambda \Sigma_{N-1} - \Sigma)^{-1} \\ &= (\lambda P_{N-1}^{-1} + \tilde{\phi}_N \tilde{\phi}'_N - \Sigma)^{-1} \\ &= (\lambda P_{N-1}^{-1} + [\tilde{\phi}_N \quad -I] \begin{bmatrix} \tilde{\phi}'_N \\ \Sigma \end{bmatrix})^{-1}. \end{aligned}$$

By the matrix inversion lemma

$$\begin{aligned} P_N &= \frac{P_{N-1}}{\lambda} - \frac{P_{N-1}}{\lambda} [\tilde{\phi}_N \quad -I] (I \\ &\quad + \begin{bmatrix} \tilde{\phi}'_N \\ \Sigma \end{bmatrix} \frac{P_{N-1}}{\lambda} [\tilde{\phi}_N \quad -I])^{-1} \begin{bmatrix} \tilde{\phi}'_N \\ \Sigma \end{bmatrix} \frac{P_{N-1}}{\lambda}. \end{aligned}$$

Moreover,

$$\begin{aligned} \tilde{\Phi}'_N W_N \tilde{Y}_N - B_N &= \lambda \tilde{\Phi}'_{N-1} W_{N-1} \tilde{Y}_{N-1} + \tilde{\phi}'_N \tilde{y}_N - \lambda B_{N-1} - B \\ &= \lambda (\tilde{\Phi}'_{N-1} W_{N-1} \tilde{Y}_{N-1} - B_{N-1}) + [\tilde{\phi}_N \quad -I] \begin{bmatrix} \tilde{y}_N \\ B \end{bmatrix}. \end{aligned}$$

Define $K_N = P_N [\tilde{\phi}_N \quad -I]$. Then,

$$\begin{aligned} K_N &= P_N [\tilde{\phi}_N \quad -I] \\ &= \frac{P_{N-1}}{\lambda} [\tilde{\phi}_N \quad -I] (I - (I \\ &\quad + \begin{bmatrix} \tilde{\phi}'_N \\ \Sigma \end{bmatrix} \frac{P_{N-1}}{\lambda} [\tilde{\phi}_N \quad -I])^{-1} \begin{bmatrix} \tilde{\phi}'_N \\ \Sigma \end{bmatrix} \frac{P_{N-1}}{\lambda} [\tilde{\phi}_N \quad -I]) \\ &= \frac{P_{N-1}}{\lambda} [\tilde{\phi}_N \quad -I] (I + \begin{bmatrix} \tilde{\phi}'_N \\ \Sigma \end{bmatrix} \frac{P_{N-1}}{\lambda} [\tilde{\phi}_N \quad -I])^{-1}. \end{aligned}$$

$$\text{It follows that } P_N = \frac{P_{N-1}}{\lambda} - K_N \begin{bmatrix} \tilde{\phi}'_N \\ \Sigma \end{bmatrix} \frac{P_{N-1}}{\lambda}.$$

Finally,

$$\begin{aligned} \theta_N &= P_N (\tilde{\Phi}'_N \tilde{Y}_N - B_N) \\ &= P_N (\lambda (\tilde{\Phi}'_{N-1} \tilde{Y}_{N-1} - B_{N-1}) + [\tilde{\phi}_N \quad -I] \begin{bmatrix} \tilde{y}_N \\ B \end{bmatrix}) \\ &= (\frac{P_{N-1}}{\lambda} - K_N \begin{bmatrix} \tilde{\phi}'_N \\ \Sigma \end{bmatrix} \frac{P_{N-1}}{\lambda}) \lambda (\tilde{\Phi}'_{N-1} \tilde{Y}_{N-1} - B_{N-1}) \\ &\quad + P_N [\tilde{\phi}_N \quad -I] \begin{bmatrix} \tilde{y}_N \\ B \end{bmatrix} \\ &= (P_{N-1} - K_N \begin{bmatrix} \tilde{\phi}'_N \\ \Sigma \end{bmatrix} P_{N-1}) (\tilde{\Phi}'_{N-1} \tilde{Y}_{N-1} - B_{N-1}) \\ &\quad + K_N \begin{bmatrix} \tilde{y}_N \\ B \end{bmatrix} \\ &= \theta_{N-1} - K_N \begin{bmatrix} \tilde{\phi}'_N \\ \Sigma \end{bmatrix} \theta_{N-1} + K_N \frac{1}{\lambda} \begin{bmatrix} \tilde{y}_N \\ B \end{bmatrix} \\ &= \theta_{N-1} + K_N (\begin{bmatrix} \tilde{y}_N \\ B \end{bmatrix} - \begin{bmatrix} \tilde{\phi}'_N \\ \Sigma \end{bmatrix} \theta_{N-1}). \end{aligned}$$

This completes the proof.

7 Diagnosis of stator winding faults

7.1 Inter-turn fault

Fig. 8 shows rotor speed trajectories for a six-step controlled motor when an inter-turn fault happens in phase a at $t = 2$ s with fault bypass resistance $R_f = 100 \Omega$. It is noted that when a fault happens the closed-loop regulation has difficulty in maintaining the required rotor speed if the leakage insulation is close to a short circuit. We will present a new detection algorithm which can detect such a fault with accuracy.

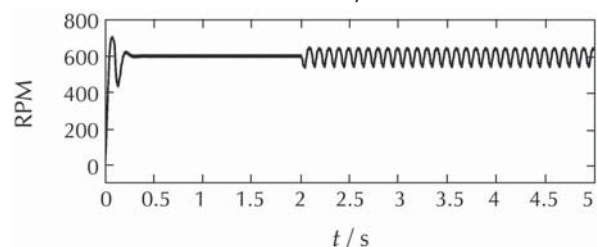


Fig. 8 Six-step controlled motor with a winding fault at $t = 2$ s, $R_f = 100 \Omega$.

7.2 Estimation of κ

The estimation algorithms under normal operating conditions provide nominal values of balanced stator winding parameters. In this section, we concentrate on

fault detection. Fault detection methods for multi-phase electrical motors can take advantage of balanced phase designs. Since all phases are symmetric, faults will alter parameter values and create an imbalanced condition between any pair of phases that can be used for detecting and isolating faults. Stator winding faults can spread quickly. Without prompt detection and protective actions, the condition can deteriorate rapidly. As a result, it is extremely important that fault detection is fast, which creates a challenging situation for designing identification algorithms. Rotor speed fluctuations can be affected by both faults and load variations. As a result, fault detection and isolation from rotor speed fluctuations are not reliable in the majority of practical situations.

In this section, we derive a new fault detection method on the basis of the parameter κ introduced in Section 2. By identifying changes in κ values, we can not only detect fault occurrence, but also identify the faulty phase and obtain the faulty current. In practical applications, it is important to know which phase is faulted. Knowing the faulty phase, one possible remedy action is to disconnect the faulty winding immediately to avoid the high shorty current from damaging other healthy windings. It is possible to control motors with only two stator windings, but this topic is beyond the scope of this paper. For fault detection, we assume that R , L , M , and λ_M are either known or estimated. Identification of κ is based on (7) $v = Ri - \kappa G_1 v_a + H \frac{di}{dt} - \kappa G_2 \frac{dv_a}{dt} + g$ and its discretized version (8). (8) can be written equivalent in a regression expression as

$$z_k = \psi'_k \kappa, \quad (18)$$

where z_k and ψ_k can be easily derived from (8). For computation of z_k and ψ_k , we point out that the inverse of H can be explicitly computed as

$$H^{-1} = \begin{bmatrix} \frac{L^2 - M^2}{L^3 - 3M^2L + 2M^3} & \frac{M^2 - LM}{L^3 - 3M^2L + 2M^3} & \frac{M^2 - LM}{L^3 - 3M^2L + 2M^3} \\ \frac{M^2 - LM}{L^3 - 3M^2L + 2M^3} & \frac{L^2 - M^2}{L^3 - 3M^2L + 2M^3} & \frac{M^2 - LM}{L^3 - 3M^2L + 2M^3} \\ \frac{M^2 - LM}{L^3 - 3M^2L + 2M^3} & \frac{M^2 - LM}{L^3 - 3M^2L + 2M^3} & \frac{L^2 - M^2}{L^3 - 3M^2L + 2M^3} \end{bmatrix}.$$

It is apparent that all previous algorithms remain viable, with y_k replaced by z_k and ϕ_k by ψ_k . As a result, we will not spell out the details here. To distinguish from the previous expressions, we will express the bias correc-

tion algorithm as

$$\kappa_N = \frac{\frac{1}{N} \tilde{\Psi}'_N \tilde{Z}_N - b}{\frac{1}{N} \tilde{\Psi}'_N \tilde{\Psi}_N - \xi}. \quad (19)$$

Note that the correction terms ξ and b are scalars, and the inverse is changed to a division here.

Example 9 We first examine the bias from measurement noises. From the regressor expression in (18), the voltage measurement noises will cause estimation bias. We evaluate estimation biases on κ by applying i.i.d. Gaussian measurement noises of zero mean but different variances. σ_v is the standard deviation of the voltage measurement noise, and σ_i is the standard deviation of the current measurement noise. Table 2 illustrates estimation errors when noise variances increase. The sampling interval is 10 ms, the estimation data length is 10000, $\mu = 0.5$ (50% inter-turn fault), $R_f = 10$, $L = 0.0064$, $M = -0.0021$, $R = 2.8750$. Apparently, the estimation biases are quite significant.

Table 2 Estimation errors on κ without bias correction.

σ_v	σ_i	True κ	Estimated κ
0.1	0.1	0.0233	0.0232
1	0.1	0.0233	0.0152
1	1.9	0.0233	0.0157

Example 10 In comparison, if the bias-corrected estimation algorithm (19) is applied, the estimation accuracy can be significantly improved. This is shown in Table 3 under the same simulation conditions. Since the noises are i.i.d., $b = 0$. Hence, the bias correction is based on ξ .

Table 3 Estimation errors on κ with bias correction.

σ_v	σ_i	True κ	Estimated κ
0.1	0.1	0.0233	0.0234
1	0.1	0.0233	0.0232
1	1.9	0.0233	0.0241

7.3 Fast fault detection with forgetting factor

One critical requirement for fast fault detection is to make the identification algorithms rely more heavily on the recent data. As discussed in Section 6, this can be achieved by employing forgetting factors. To illustrate the impact of forgetting factors on the speed of fault detection, we select different values of λ and show the corresponding trajectories of estimates in tracking κ af-

ter a fault occurrence in Fig. 9. It is clear that to achieve fast tracking capability, a relatively small λ should be selected.

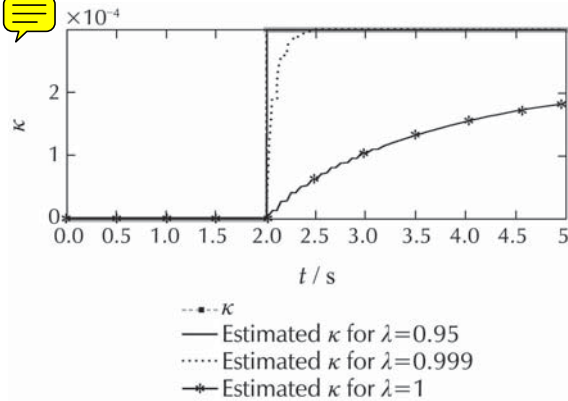


Fig. 9 Estimation of κ under different forgetting factor λ .

7.4 Statistics for ξ and b

The bias correction algorithm (19) relies on the knowledge of ξ and b to devise correction actions. In practical applications, such covariance values may not be available *a priori*. As a result, they need to be estimated also. We now present an estimation scheme for ξ and b . For simplicity, we assume that all sensor noises are Gaussian i.i.d. random variables.

In our study, the motor starts without winding fault. This implies that at the starting time, $\kappa = 0$. Consequently, under normal operation, $Z_N = 0$. From the expression $\frac{1}{N}\Psi'_N Z_N \rightarrow A$, we conclude that $A = 0$ if $\kappa = 0$. From the equation $\frac{1}{N}\tilde{\Psi}'_N \tilde{Z}_N \rightarrow A + b$, we obtain $\frac{1}{N}\tilde{\Psi}'_N \tilde{Z}_N \rightarrow b$. This relationship can be used for b determination when we know that no fault is present.

The identification equation for κ is $Z_N = \Psi_N \kappa$. We assume that the three-phase motor model is known. Hence Ψ_N is known. The measurement equations are $\tilde{z}_k = z_k + e_k$; $\tilde{i}_k = i_k + \varepsilon_k$. Substituting these equations in Ψ_N , we obtain $\tilde{\Psi}_N = \Psi_N + \Delta_N$. The bias correction term is the limit $\frac{1}{N}\Delta'_N \Delta_N \rightarrow \xi$, which can be used to estimate ξ .

8 Conclusions

Parameter identification for three-phase motors is a difficult task due to measurement noise. The methodology introduced in this paper enhances the traditional

LS algorithms with integrated bias removal and forgetting factors. By relating winding faults to changes in a characterizing variable, fault detection is explored in the identification framework. In addition, by exploiting the symmetry of phases in balanced PM stator windings, we introduce a ratio test to isolate faults with fast response and convergence. We demonstrate that our bias removal algorithms can significantly improve fault detection reliability when measurement noises are present. A related topic is control algorithms for running a three-phase motor when only two phases are functional. Combined with our fault detection algorithms, this joint diagnosis and control strategy can potentially provide robustness in motor operations when faults occur.

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